THE PLAN

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Objective

Build myself into a **Data Scientist capable of working effectively in the Financial Crime domain** over the next 8–9 months.

The target is to become a Data Scientist who can:

- Understand Financial Crime problems
- Work with banking and transaction data
- Use Python extensively
- Write advanced SQL
- Perform statistical and exploratory analysis
- Build fraud and AML detection systems
- Engineer behavioural and risk features
- Build and evaluate machine-learning models
- Apply graph/network analytics to Financial Crime
- Understand and experiment with GNNs and temporal graph models
- Work with large-scale data using Spark
- Understand real-time detection architectures
- Communicate technical findings in a Financial Crime context

Target organisations

- National and international banks
- NBFCs
- Established fintechs and financial startups
- Payment companies
- Government / financial institutions
- Organizations such as NPCI, RBI and similar institutions

The goal is NOT:

> Become a Financial Crime domain expert in 9 months.

The goal IS:

> Become a strong Data Scientist who understands the Financial Crime domain well enough to identify problems, work with financial data, build analytical and machine-learning solutions, and collaborate effectively with domain experts.

Guiding Philosophy

This is a **project-first learning journey**.

I don't want to spend months learning theory in isolation.

The approach is:

> **Learn → Build → Break → Investigate → Improve → Document**

Each project should introduce a new layer of knowledge while reusing what was learned previously.

The Big Project — Alpha Capital Bank

The entire journey takes place inside one fictional financial institution:

Alpha Capital Bank

I am joining Alpha Capital Bank as a:

> **Data Scientist — Financial Crime Analytics**

The bank is gradually building its Financial Crime analytics capability.

Each project represents a problem the bank has asked me to solve.

The projects are independent enough to stand on their own as portfolio pieces, but they share:

- The same bank
- The same customers
- The same accounts
- The same transaction history
- The same entities
- The same Financial Crime scenarios
- The same evolving data platform

Therefore:

> **The portfolio should look like one large Financial Crime Data Science project when viewed from a distance.**

Overall Progression

```
                     ALPHA CAPITAL BANK
                             |
                             ▼
                 PROJECT 0 — DATA FOUNDATION
                             |
                             ▼
                PROJECT 1 — CUSTOMER INTELLIGENCE
                             |
                             ▼
```

```
              PROJECT 2 — TRANSACTION MONITORING
                            │
                            ▼
               PROJECT 3 — RISK SCORING
                            │
                            ▼
                PROJECT 4 — FRAUD DETECTION
                            │
                            ▼
                  PROJECT 5 — AML MACHINE LEARNING
                            │
                            ▼
          PROJECT 6 — GRAPH & NETWORK INTELLIGENCE
                            │
                   ┌────────┴────────┐
                   ▼                 ▼
             Graph Analytics      Graph ML
                   │                 │
                   └────────┬────────┘
                            ▼
                   Temporal Graphs
                            │
                            ▼
                     Temporal GNNs
                            │
                            ▼
              PROJECT 7 — BIG DATA PLATFORM
                            │
                            ▼
               PROJECT 8 — REAL-TIME DETECTION
                            │
                            ▼
               PROJECT 9 — END-TO-END CAPSTONE
```

Roadmap Structure

Project	Duration	Level	Primary Focus
Project 0	4 weeks	Foundation	Banking data architecture
Project 1	4 weeks	Basic	Customer & behavioural analytics
Project 2	4 weeks	Basic	Transaction monitoring

Project	Duration	Level	Primary Focus
Project 3	4 weeks	Basic → Intermediate	Financial Crime risk scoring
Project 4	4 weeks	Intermediate	Fraud ML
Project 5	4 weeks	Intermediate	AML ML & anomaly detection
Project 6	5 weeks	Advanced	Graph, network & temporal intelligence
Project 7	4 weeks	Advanced	Big Data / Spark
Project 8	4 weeks	Advanced	Real-time detection
Project 9	3–4 weeks	Capstone	End-to-end platform

Total: ~40 weeks / 9 months

If necessary, Projects 8–9 can be compressed to bring the roadmap closer to 8 months.

The Four Core Skill Tracks

These tracks run throughout the entire roadmap.

Track 1 — Python

Foundation

- Python
- Functions
- Data structures
- File handling
- pandas
- NumPy

Intermediate

- Feature engineering
- Statistical analysis
- scikit-learn
- Pipelines
- Modular programming
- Testing
- Logging

Advanced

- PySpark
- Distributed processing
- Graph processing
- Streaming
- Production-oriented code

Final target

> **Python should become the primary language I use for Data Science.**

Track 2 — SQL

SQL is a **core competency**, not a side skill.

Foundation

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- CASE
- JOINs

- Aggregations

Intermediate

- CTEs
- Subqueries
- Window functions
- Date/time operations
- Rolling metrics
- Analytical SQL

Advanced

- Complex behavioural features
- Temporal analysis
- Query optimization
- Execution plans
- Large-scale SQL
- Spark SQL

Final target

> I should be comfortable receiving a messy Financial Crime data problem and using SQL to construct the analytical dataset myself.

Track 3 : Statistics & Analytics

Progression:

```
Descriptive Statistics
        ↓
Distributions & Outliers
        ↓
Behavioural Analytics
        ↓
Anomaly Detection
```

```
        ↓
Probability & Statistical Testing
        ↓
Classification
        ↓
Imbalanced Learning
        ↓
Threshold Optimization
        ↓
Calibration
        ↓
Model Monitoring
```

Track 4 : Financial Crime Domain

Domain knowledge progresses alongside the technical work.

Foundation

- Banking ecosystem
- Accounts
- Customers
- Transactions
- KYC
- AML
- Fraud

Intermediate

- Transaction monitoring
- Suspicious activity
- Fraud typologies
- AML typologies
- Customer risk
- Alerts
- False positives

Advanced

- Mule networks
- Fraud rings
- Network-based crime
- Behavioural risk
- Graph-based detection
- Real-time detection
- Model governance
- Investigation workflows

Skill Priority

Not every skill has equal importance.

Tier 1 — Must Become Strong

Python

★★★★★

SQL

★★★★★

Statistics / Analytics

★★★★☆

Machine Learning

★★★★☆

Financial Crime fundamentals

★★★★☆

Tier 2 — Must Have Hands-On Experience

PySpark

★★★★☆

Databricks

★★★☆☆

Graph Analytics

★★★★☆

Network Analysis

★★★★☆

Data Engineering

★★★☆☆

Tier 3 — Advanced Differentiators

GNNs

★★★☆☆

GCN / GraphSAGE / GAT

★★★☆☆

Temporal Graphs

★★★☆☆

Temporal GNNs

★★☆☆☆

Kafka / Streaming

★★★☆☆

The objective isn't mastery of every Tier 3 technology.

The objective is to have **real hands-on exposure and understand where these techniques fit.**

What I Will Not Optimize For

I am not trying to become:

- A regulatory compliance specialist
- An AML investigator
- A pure ML researcher
- A graph research scientist
- A data engineer
- A quantitative researcher

The primary identity remains:

> **Data Scientist**

Financial Crime is the domain in which I am applying Data Science.

What Every Project Must Demonstrate

Every project should contain some combination of:

1. Domain understanding

What Financial Crime problem are we solving?

2. Data

What data is required?

3. SQL

What can be extracted or engineered using SQL?

4. Python

What analysis or modelling is performed in Python?

5. Statistics

What statistical reasoning is required?

6. Modelling

Does machine learning improve the solution?

7. Evaluation

How do we measure success?

8. Engineering

How would this work at scale?

9. Business reasoning

Would a bank actually find this useful?

10. Limitations

What could go wrong?

Final Target Skill Profile

By the end of the journey:

Area	Target
Python	Strong
SQL	Strong / Advanced
Statistics	Strong working knowledge
Machine Learning	Strong working knowledge

Area	Target
Fraud Analytics	Strong working knowledge
AML Analytics	Strong working knowledge
Transaction Monitoring	Strong working knowledge
Risk Scoring	Strong working knowledge
Graph Analytics	Hands-on
Fraud Ring Detection	Hands-on
Mule Detection	Hands-on
Graph ML	Working knowledge + project experience
GCN / GraphSAGE / GAT	Working knowledge
Temporal Graphs	Working knowledge
Temporal GNNs	Exposure + experimental project
PySpark	Hands-on
Databricks	Hands-on exposure
Kafka	Hands-on exposure
Streaming	Working knowledge
Data Engineering	Working knowledge
Model Monitoring	Working knowledge

End State

At the end of 8–9 months, I should be able to look at a Financial Crime Data Scientist job description and understand the majority of the technical concepts being discussed.

I should be able to:

- Work comfortably with Python
- Write advanced SQL
- Analyze large transaction datasets
- Engineer behavioural features

- Build transaction-monitoring rules
- Build fraud models
- Build AML models
- Handle imbalanced datasets
- Perform anomaly detection
- Understand false positives and investigator workload
- Build customer risk scores
- Detect fraud rings
- Detect potential mule networks
- Construct financial graphs
- Engineer graph-based risk signals
- Apply graph ML
- Experiment with GCN/GraphSAGE/GAT
- Understand temporal graphs
- Experiment with temporal GNNs
- Process large datasets with Spark
- Understand Kafka and streaming architectures
- Think about real-time Financial Crime detection
- Connect all these components into a coherent system